

BAHub - Phase 9: Testing & Requirement Traceability Matrix (RTM)

This document contains UAT scenarios, positive/negative test cases, and the Requirement Traceability Matrix (RTM) for the **BAHub** workspace application.

1. User Acceptance Testing (UAT) Scenarios

UAT Scenario 1: Collaborative Discovery Workshop

- **Goal:** Verify that a BA can map stakeholders, create requirements with sequential IDs, decompose them into user stories, and verify that typing indicators work.
- **Steps:**
 1. The BA registers a new organization and invites two team members.
 2. The BA creates Project Alpha, registers stakeholders, and assigns influence ratings.
 3. The BA and a developer open the Requirements Split Grid. The developer verifies that presence indicators appear for both users.
 4. The BA inputs five requirements and verifies that sequential IDs (REQ-001 through REQ-005) generate correctly.
 5. The BA decomposes REQ-001 into three User Stories and verifies they appear on the Agile Backlog board.

UAT Scenario 2: Document Baseline & Sign-off Review

- **Goal:** Verify the compilation of project assets and the Product Owner sign-off flow.
 - **Steps:**
 1. The BA opens the Document Generator and compiles a new BRD.
 2. The BA verifies that compiled requirements, stakeholders, and stories format correctly in the Markdown preview pane.
 3. The BA transitions the document status to REVIEW .
 4. The PO logs in, reviews the document, and clicks the "Sign Off" button.
 5. The system locks the document, sets the status to SIGNED_OFF , and records the signature details.
 6. The BA downloads the signed document as a PDF.
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2. Test Cases (Functional, Positive, & Negative)

2.1 Positive Test Cases (Normal Operation)

Test Case ID	Component	Description	Input Conditions	Expected Result	Status
TC-POS-001	Authentication	User logs in using valid credentials.	Username: analyst , Password: AnalystP@ss123	JWT access and refresh tokens returned; user redirected to dashboard.	Pass
TC-POS-002	Requirements	Create requirement.	Title: Core UI Grid , Priority: HIGH , Project: Project Alpha	Database count increments, and a requirement is saved with ID REQ-001 .	Pass
TC-POS-003	Document Compile	Export project document as PDF.	Click "Download PDF" on the BRD Compiler view.	PDF downloads in under 5 seconds with correct formatting.	Pass
TC-POS-004	Jira Integration	Synchronize backlog stories to Jira.	Valid Jira integration credentials saved; click "Sync Backlog".	Celery task completes. UserStory records updated with jira_key and jira_url .	Pass

2.2 Negative Test Cases (Robustness & Security)

Test Case ID	Component	Description	Input Conditions	Expected Result	Status
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TC-NEG-001	Multi-Tenancy	Access requirement from another organization via URL.	User logged in to Org X attempts to call GET <code>/api/v1/requirements/<uuid></code> for Org Y.	API returns <code>404 Not Found</code> or <code>403 Forbidden</code> JSON envelope.	Pass
TC-NEG-002	Requirements	Attempt to save requirement with missing fields.	Submit requirement form with Title blank.	API returns <code>400 Bad Request</code> with field validation errors.	Pass
TC-NEG-003	Billing	Exceed seats limit on a free plan tier.	Invite a 6th user to an organization on the default free tier.	invitation blocked with an error message: "Seats limit exceeded. Upgrade to Pro."	Pass
TC-NEG-004	Integrations	Sync backlog with invalid Jira API token.	Set an invalid token in Integrations configuration and run sync.	Sync log status set to <code>FAILED</code> with error message in history logs.	Pass

3. Requirement Traceability Matrix (RTM)

The matrix below maps Business Requirements down to Functional Requirements, Technical Specifications, and their corresponding validation Test Cases.

Business Requirement (BR)	Functional Requirement (FR)	Tech Spec (SRS)	Verification Test Cases	Status
BR-101: Secure Multi-Tenancy	FUNC-002: Project Scoping & Member RBAC	SRS Section 4.2: Database Isolation	TC-NEG-001 (Multi-Tenancy check)	Verified
BR-102: Requirements Editor	FUNC-004: Notion Requirements Split Editor	SRS Section 2.2: Requirements API	TC-P0S-002 , TC-NEG-002	Verified
BR-103: Stakeholder Registry	FUNC-003: Stakeholder Matrix Grid	SRS Section 2.2: Stakeholders API	TC-P0S-002 (Traceability mapping)	Verified
BR-104: Backlog Sync	FUNC-005: Agile Backlog Board & User Story	SRS Section 2.3: JIRA Sync API	TC-P0S-004 , TC-NEG-004	Verified
BR-105: Document Compiler	FUNC-011: Document Publisher & PDF Compiler	SRS Section 2.3: Documents API	TC-P0S-003	Verified